# Dynamic Token Augmentation Mamba for Cross-Scene Classification of Hyperspectral Image

Xizeng Huang, Yuxiang Zhang, *Senior Member, IEEE*, Fulin Luo, *Senior Member, IEEE*, and Yanni Dong, *Senior Member, IEEE*

***Abstract*— Cross-scene classification of hyperspectral image (HSI) based on single-source domain generalization (SDG) focuses on developing a model that can effectively classify images from unseen target domains using only source domain images, without the need for retraining. Most existing SDG approaches for cross-scene classification rely on convolutional neural networks (CNNs). However, the convolutional kernel operation causes the model to emphasize local object features, which can lead to overfitting on the source domain and limits its ability to generalize. Recently, methods based on the state space model (SSM) have demonstrated excellent performance in image classification by capturing global features across different image patches. Building on this inspiration, we propose a novel approach called dynamic token augmentation mamba (DTAM), which aims to explore the potential of SSMs in the cross-scene classification of HSI. The method gradually focuses on the global features of the image by constructing hidden states for HSIs unfolded into long sequences. To further enhance the global features of HSIs, we design a dynamic token augmentation (DTA) module to transform the sample features by perturbing the contextual information while preserving the object information tokens. Additionally, we introduce a loss of classified compensation combined with labels of random samples to suppress the excessive narrowing of the feature range learned by the model. Comprehensive extensive experiments on three publicly available HSI datasets show that the proposed method outperforms the state-of-the-art (SOTA) method. Our code is available at https://github.com/Varro-pepsi/DTAM.**

***Index Terms*— Cross-scene classification, data manipulation, hyperspectral image (HSI), single-source domain generalization (SDG), state space model (SSM).**

## I. Introduction

HYPERSPECTRAL image (HSI) contains detailed spectral information in hundreds of narrow bands [1], [2]. Utilizing HSI to identify material classes offers significant advantages since materials have unique spectral reflectance characteristics [3], [4], [5]. Classification technology of HSI is widely used in agriculture [6], environmental monitoring [7], mining [8], and other fields [9]. However, variations in environmental conditions, sensor characteristics, and material properties lead to domain shifts between data from different scenes [10]. This results in models trained on data from one scene (source domain) being unable to maintain superior classification performance in a new scene (target domain) [11]. To reduce the impact of domain shifts on model performance, the cross-scene classification technique of HSI has been proposed. Currently, domain adaptation (DA) and domain generalization (DG) are the two main research ideas for this technique.

DA aims to train a model using labeled sources domain samples and a minimum number of unlabeled target domain samples to maintain high classification accuracy in unseen target domains [12]. DA methods are categorized into three main types based on how they reduce data distribution variance: discrepancy-based, adversarial-based, and reconstruction-based methods [13]. The discrepancy-based DA methods utilize different metrics to reduce feature variations across domains. Long et al. [14] minimized the domain shifts by reducing the feature distance in nuclear Hilbert space. The core idea of adversarial-based DA methods is to constrain the similarity of feature representations of different domains by adversarial training at the deep network layer. Ganin and Lempitsky [15] used adversarial learning to obfuscate the features learned by the model to ensure that the feature distributions on different domains are similar. The reconstruction-based DA methods utilize autoencoders to minimize cross-domain reconstruction errors. Glorot et al. [16] prompted the model to learn features common to both source and target domain data by reconstructing concatenated sets of data from different domains. However, whenever new TD data is acquired, the DA methods have to be returned and retrained using the new TD data. This greatly limits the model's practical deployment scenarios.

In contrast, DG aims to develop models that generalize across unseen domains without the need to acquire any TD data during training [17], [18], [19], [20], [21]. DG technique focuses on learning domain-independent features, i.e., domain-invariant features. These features are robust to changes in different scenes, thus allowing the model to perform well in new scenarios without the need for domain-specific adaptation. According to the number of source domains available during

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model training, DG can be divided into single-source DG (SDG) research direction and multiple-source DG (MDG) research direction. MDG can capture domain-invariant features from samples of multisource domains, whereas SDG is limited to learning from a single source domain. Therefore, SDG is more difficult to learn domain-invariant features than MDG. To capture more domain-invariant features from a limited number of single-source domain samples, many SDG methods take data augmentation, adversarial learning, and so on. Qiao et al. [22] used adversarial training combined with meta-learning to generate fictitious and efficient new samples, and used the Wasserstein autoencoder (WAE) to relax widely used worst-case constraints. Cugu et al. [23] proposed a method called attention consistency on visual corruptions, which alters the SD samples by visual corruptions to generate more feature-rich samples. Wang et al. [24] enhanced the generalization ability of the model by synthesizing different separate samples complementary to the source domain image.

Generally, the existing SDG works for cross-scene classification of HSI are primarily built upon convolutional neural networks (CNNs). However, the convolutional kernel operation in CNNs causes the model to emphasize local object features, which can lead to overfitting on the source domain and limits its ability to generalize [25]. To tackle this problem, we propose a new method called dynamic token augmentation mamba (DTAM) for cross-scene classification of HSI based on the state space models (SSMs). The mamba layer in the method gradually focuses on the global features of the samples by constructing hidden states for HSIs unfolded into long sequences. Additionally, we specifically designed a sequence-level dynamic token augmentation (DTA) module in the headend of the mamba layer. Specifically, the DTA module mainly consists of the object tokens selection (OTS) submodule and the context tokens perturbation (CTP) submodule. The OTS implements the preservation of tokens with object information in the original features. The CTP makes context tokens with more complex information using random sample features. The original features are implicitly augmented by reorganizing the outputs of the two submodules. Meanwhile, we utilize the loss of classified compensation to improve the robustness of the features learned by the model. The main contributions of this work are outlined below.

1) We propose a novel framework for cross-scene classification of HSI based on SSMs. The framework gradually focuses on the global features of the samples by constructing hidden states for HSIs unfolded into long sequences.
2) To further enhance the global features of HSIs, we design a DTA module to transform the sample features by perturbing the context information while preserving the object information tokens.
3) To increase the robustness of the model, we introduce a loss of classified compensation combined with labels of random samples to suppress the excessive narrowing of the feature range learned by the model.

Sections II–V of this article are organized as follows: Section II explores research on DG and mamba. Section III offers a detailed explanation of the proposed DTAM. Section IV presents experimental data and detailed results to highlight the effectiveness of our approach. Section V summarizes our findings and outlines future research directions.

## II. Related Work

### A. Single-Source DG

SDG tends to promote the capacity of the model to generalize, where training from a single source domain can achieve excellent classification accuracy on unseen domains. Due to the limitation of the number of source domains, most existing SDG approaches rely on data manipulation ways to generate new samples with richer features, thereby enhancing the capacity of the model to generalize. Li et al. [26] proposed a method to incrementally generate new samples of multiple domains by modeling various photometric and geometric transformations in unseen domains. Zhao et al. [27] constructed spectral enhancement domain and spatial enhancement domain using adaptive instance normalization (AdaIN) layer and spatial transformation network, respectively. Zhang et al. [28] used comparative and adversarial learning to generate extended domain samples and learn domain-invariant features based on an encoder-randomizer–decoder architecture. Dong et al. [29] resolved the effect of domain bias by modeling the spatial–spectral style of unseen domains and building a causal contribution discriminator. Inspired by the conversion of images into multiple frequency components, Huang et al. [30] proposed a method that performs randomization in the frequency space domain to generate new samples. However, the works mentioned above are based on CNN as the backbone network. The convolutional kernel operation causes the model to emphasize local object features, which can lead to overfitting on the source domain and limits its ability to generalize [25]. Therefore, the new approach needs to be built on the backbone network that is biased toward capturing global object features.

### B. State Space Models

Recently, the advantages of SSMs in sequence analysis have gained more and more attention from scholars [31], [32], [33]. SSMs are not only capable of long-range sequence modeling but also have linear computational complexity.

*1) Structured State Space Sequence Model (S4):* Inspired by SSM and deep learning, structured state space sequence models have obtained increasing attention in dealing with sequence data. Mathematically, SSM can usually be represented by a set of linear ordinary differential equations (ODEs) as follows:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ denotes the evolution parameter, $\mathbf{B} \in \mathbb{R}^{N\times D}$ denotes the transformation matrix, $\mathbf{C} \in \mathbb{R}^{D\times N}$ denotes the projection parameter, and $\mathbf{D} \in \mathbb{R}^1$ denotes the hopping connectivity coefficient. SSM can project the signal $x(t) \in \mathbb{R}^D$ to $y(t) \in \mathbb{R}^D$ by hidden states $h(t) \in \mathbb{R}^N$. Here $t$ denotes the time step, $D$ is the initial dimension, and $N$ is the state dimension. Since SSM is continuous in time, it cannot be

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directly integrated into deep learning. The problem is solved by the proposal of S4. S4 is a discrete version of SSM. The time parameter $\mathbf{\Delta} \in \mathbb{R}^D$ converts the continuous state parameters $\mathbf{A}$ and $\mathbf{B}$ into discrete state parameters $\bar{\mathbf{A}} \in \mathbb{R}^{N\times N}$ and $\bar{\mathbf{B}} \in \mathbb{R}^{N\times D}$. This transformation employs a zero-order holding (ZOH) method, which is expressed in the following equations:

$$\bar{\mathbf{A}} = \exp(\mathbf{\Delta A}) \tag{3}$$

$$\bar{\mathbf{B}} = (\mathbf{\Delta A})^{-1}(\exp(\mathbf{\Delta A}) - \mathbf{I}) \odot \mathbf{\Delta B}. \tag{4}$$

Therefore, (1) and (2) can be expressed after discretization as follows:

$$h(t) = \bar{\mathbf{A}}h(t-1) + \bar{\mathbf{B}}x(t) \tag{5}$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t). \tag{6}$$

Finally, S4 computes the output in deep learning through a global convolution operation as follows:

$$\bar{K} = \left(\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}\right) \tag{7}$$

$$\mathbf{y} = \mathbf{x} \star \bar{K} \tag{8}$$

where $L$ is the length of the input sequence data $\mathbf{x}$ and $\bar{K}$ is a structured SSM kernel. $\star$ denotes the convolution operation.

*2) Mamba Model:* The mamba model [34] enhances the performance of the S4 model by introducing a dynamic scanning mechanism, which is termed the Selective Scan-Based S4 (S6) model. Specifically, it introduces an adaptive parameterization mechanism for $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{\Delta}$, which is formulated as follows:

$$\mathbf{B} = \mathrm{P_B}(\mathbf{x}) \tag{9}$$

$$\mathbf{C} = \mathrm{P_C}(\mathbf{x}) \tag{10}$$

$$\mathbf{\Delta} = \tau_\Delta(Parameter + \mathrm{P}_\Delta(\mathbf{x})) \tag{11}$$

where $\mathbf{x} \in \mathbb{R}^{B\times L\times D}$ is the batch input sequence data. Here $B$ denotes the input batch size, $L$ denotes the sequence data length, and $D$ is the sequence data dimension. The functions $\mathrm{P_B}(\cdot)$ and $\mathrm{P_C}(\cdot)$ linearly project the input $\mathbf{x}$ into $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$ and $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$. Furthermore, the function $\mathrm{P}_\Delta(\cdot)$ converts to size $\mathbb{R}^{B\times L\times 1}$, which is then broadcast into dimension $D$. Parameter is a randomly selected set of extremely small values. $\tau_\Delta(\cdot)$ is typically a softplus activation function that produces a new $\mathbf{\Delta}$ of size $\mathbb{R}^{B\times L\times D}$.

Mamba [34] further optimizes its long-range relational modeling capability by introducing a selectivity mechanism. Currently, many works explore the potential of mamba in computer vision methods. The method [35] labels image sequences with positional embedding and represents the SSM with bidirectional compressed vision. Li et al. [36] propose a generalized design for extending the mamba architecture to arbitrary multidimensional data, enabling alternate deconstruction of input data in different dimensions. Furthermore, Huang et al. [37] introduce a novel local scanning approach to effectively capture local dependencies while maintaining the global view of mamba. Although the above work demonstrates the strong performance of mamba for processing long sequences, no scholar has yet exploited its generalization potential in the cross-scene classification of HSI.

## III. Proposed Method

### A. Framework of the Network

For the cross-scene classification task of HSI, each input data to the network is represented as $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^B \in \mathbb{R}^{B\times p\times p\times C}$, which $\mathbf{x}$ in $\mathbf{X}$ is a 3-D patch of size $p\times p\times C$. The class label corresponding to $\mathbf{X}$ is denoted as $\mathbf{Y} = \{\mathbf{y}_i\}_{i=1}^B$. Here $C$, $p$, and $B$ denote the spectral dimension of the sample, the spatial dimension of the sample, and the amount of samples in each batch fed into the network, respectively. The flowchart of the proposed DTAM method is shown in Fig. 1. First, the input data is sequentially unfolded into a 2-D patch. Subsequently, the feature $\mathbf{Z} = \{\mathbf{z}_i\}_{i=1}^B \in \mathbb{R}^{B\times N\times D}$ of the input data is obtained by adding positional encoding and spectral dimension projection through the patch embedding layer. Here, $N$ is equal to the square of $p$ and $D$ denotes the size of the dimension after the projection of the spectral dimension. Then the features are implicitly augmented using a DTA module and further global features are extracted using the mamba layer. The predicted probability $\{\mathbf{p}_i\}_{i=1}^B$ of the input data is obtained through the classifier header after N times of augmentations and mamba layers. Finally, the loss of classified compensation $\mathcal{L}_{\mathrm{CC}}$ is computed.

### B. Dynamic Token Augmentation

To further enhance the global features of HSIs, we designed a token-level enhancement module called DTA. We consider that the HSI sample features are composed of features that contain object information and features that contain context information. For example, in the sample whose class is buildings, some tokens contain object information about buildings, and the rest contain context information about other classes such as grass and roads. Due to the different collection times and regions of the images, the context information of the same class of samples may change more obviously, but the same features will be preserved in the object information corresponding to the class. Second, due to the influence of lighting conditions, climate, and other conditions at the time of collection, the spectral reflectance characteristics of the same class on HSIs of different scenes are inconsistent. Therefore, while improving the model's focus on features that contain object information, the influence of differences on spectral reflectance characteristics should also be reduced. For this purpose, we designed the DTA. As shown in the right part of Fig. 1, the DTA mainly consists of the OTS submodule, CTP, and token reorganization submodule. Given the features $\{\mathbf{z}_i\}_{i=1}^B$ of the input data, we shuffle the batch sequence to generate random features $\{\mathbf{z}_j\}_{j=1}^{B^s}$, where $B^s$ denotes the new batch sequence. In the subsequent token augmentation, the sequence of batches of random features remains the same as that sequence. We then input the original features of the input data into the OTS submodule and the random features into the CTP submodule.

*1) Object Tokens Selection:* The OTS submodule is designed to filter the tokens that contain object information. First, we use the token space relational search component to produce the map of the original feature token space relations. Our motivation comes from [38]. Tokens that have high

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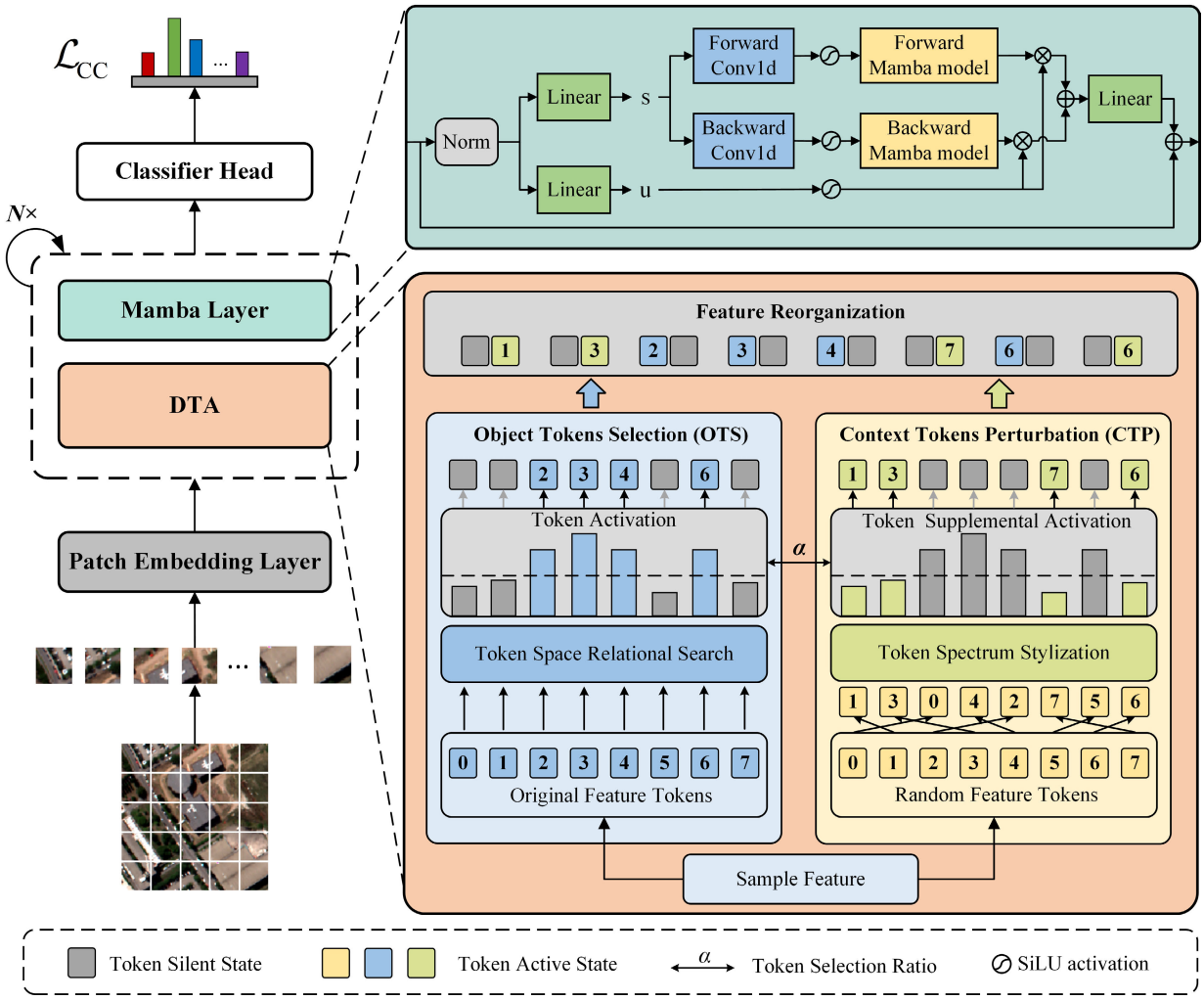



Fig. 1. Flowchart of the proposed DTAM.

spatial associations with other tokens in the feature may contain object information. It uses the self-attention mechanism Attention(·) [39] and multilayer perceptions MLP(·) [40] to search the spatial relationship between tokens. During the process of image feature extraction by the network, the locations of tokens containing object information are relatively stable. Therefore, we set up a memory attention map $\mathbf{M}_{\text{memory}}$ to mitigate the drastic fluctuations in the spatial relational search. The memory attention map is updated after each search for spatial relationships between feature tokens. Given a feature $\mathbf{z}_i$, the process of token space relationship map $\mathbf{M}_i \in \mathbb{R}^{N\times D}$ making is formulated as follows:

$$\mathbf{M}_i = \text{MLP}(\text{Attention}(\mathbf{z}_i)) + \mathbf{M}_{\text{memory}}. \tag{12}$$

On this basis, we compute the activation value in the token space relationship map which is formulated as follows:

$$E(\mathbf{M}_i)_n = \frac{1}{D}\sum_{d=0}^{D-1}\mathbf{M}_i(n,d) \tag{13}$$

where $E(\mathbf{M}_i)_n$ denotes the activation value of the $n$th token in the spatial relationship map. By calculating all the tokens we get the map $E(\mathbf{M}_i)$ of spatial relationship activation values. The larger the activation value, the more the token contains object information. Therefore, we perform a thresholding operation to obtain the object information activation mask $\mathbf{M}_i^{\text{act}} \in \mathbb{R}^{N\times D}$. Specifically, we compute the $\alpha$-th percentile as the threshold, denoted as $q_\alpha$. If the activation value in $E(\mathbf{M}_i)$ is greater than the threshold, the element is set to 1, and set to 0 otherwise. It is formulated as follows:

$$\mathbf{M}_i^{\text{act}}(n,d) = \begin{cases} 1, & E(\mathbf{M}_i)_n \geq q_\alpha \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $\alpha$ is the adjustable token selection ratio. Finally, we could extract the object tokens $\mathbf{z}_{\text{OTS}}$ by filtering from the original feature $\mathbf{z}_i$ by using the binary mask $\mathbf{M}_i^{\text{act}}$, which is formulated as follows:

$$\mathbf{z}_{\text{OTS}} = \mathbf{M}_i^{\text{act}} \odot \mathbf{z}_i. \tag{15}$$

*2) Context Tokens Perturbation:* The CTP submodule is designed to generate new tokens with richer spectral features. During HSI acquisition, differences in environmental conditions can lead to differences in the spectral features of the same material in the source and target domains. Therefore, we need to minimize the difference in spectral features between the two domains. However, lacking the guidance of target domain

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samples, if the spectral information of all original features is directly perturbed, it may cause the loss of object information corresponding to the class. Therefore, we take a conservative idea by disturbing only the spectral information of the context tokens. Specifically, given a random feature $\mathbf{z}_j$, we first use the Shuffle$(\cdot)$ function to break up the spatial relationships between feature tokens from the token level

$$\mathbf{z}_j^s = \text{Shuffle}\left(\mathbf{z}_j\right). \tag{16}$$

On this basis, we perturb the spectral information of $\mathbf{z}_j^s$ by the token spectrum stylization component. We primarily implement the process using AdaIN [41], which is defined as follows:

$$\text{AdaIN}(\mathbf{z}, \mathbf{n}) = \text{FC}_1(\mathbf{n})\frac{\mathbf{z} - \mu(\mathbf{z})}{\sigma(\mathbf{z})} + \text{FC}_2(\mathbf{n}). \tag{17}$$

Here $\mathbf{z}$, $\mathbf{n}$, and FC$(\cdot)$ denote sample features, random noise $\mathbf{n} \sim \mathbf{N}(\mathbf{0}, \mathbf{1})$, and fully connected layer, respectively. The operation of the token spectrum stylization component is as follows. We compute the style representation of $\mathbf{z}_j^s$, which is the channel mean $\mu(\mathbf{z}_j^s)$ and standard deviation $\sigma(\mathbf{z}_j^s)$. The formula for calculating the style representation is as follows:

$$\mu\left(\mathbf{z}_j^s\right) = \frac{1}{N}\sum_{n=1}^{N}\mathbf{z}_j^s \tag{18}$$

$$\sigma\left(\mathbf{z}_j^s\right) = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{z}_j^s - \mu\left(\mathbf{z}_j^s\right)\right)^2} \tag{19}$$

where $N$ denotes the number of tokens in $\mathbf{z}_j^s$. Then $\mu(\bar{\mathbf{z}}_j^s)$ and $\sigma(\bar{\mathbf{z}}_j^s)$ corresponding to $\bar{\mathbf{z}}_j^s$ are randomly selected and adaptively linearly combined with $\mu(\mathbf{z}_j^s)$ and $\sigma(\mathbf{z}_j^s)$ to obtain $\hat{\mu}$ and $\hat{\sigma}$. This process is formulated as follows:

$$\hat{\mu} = \lambda\mu\left(\mathbf{z}_j^s\right) + (1-\lambda)\mu\left(\bar{\mathbf{z}}_j^s\right) \tag{20}$$

$$\hat{\sigma} = \lambda\sigma\left(\mathbf{z}_j^s\right) + (1-\lambda)\sigma\left(\bar{\mathbf{z}}_j^s\right) \tag{21}$$

where $\lambda$ is the adaptive learning parameter. After this, we replace the mean and standard deviation of the noise in (17) with $\hat{\mu}$ and $\hat{\sigma}$ to stylize the token spectrum

$$\hat{\mathbf{z}}_j^s = \hat{\mu}\frac{\mathbf{z}_j^s - \mu\left(\mathbf{z}_j^s\right)}{\sigma\left(\mathbf{z}_j^s\right)} + \hat{\sigma}. \tag{22}$$

Following this, we perform supplemental activation of $\hat{\mathbf{z}}_j^s$ against the activation mask $\mathbf{M}_i^{\text{act}}$ of the original feature $\mathbf{z}_i$. The activation mask $\mathbf{M}_j^{\text{act}} \in \mathbb{R}^{N\times D}$ for $\hat{\mathbf{z}}_j^s$ is formulated as follows:

$$\mathbf{M}_j^{\text{act}}(n,d) = \begin{cases} 0, & \mathbf{M}_i^{\text{act}}(n,d) \\ 1, & \text{otherwise} \end{cases}. \tag{23}$$

Finally, we could obtain new context tokens $\mathbf{z}_{\text{CTP}}$ by using the binary mask $\mathbf{M}_j^{\text{act}}$, which is formulated as follows:

$$\mathbf{z}_{\text{CTP}} = \mathbf{M}_j^{\text{act}} \odot \hat{\mathbf{z}}_j^s. \tag{24}$$

*3) Feature Reorganization:* This submodule implements the reorganization of the feature using the obtained object tokens $\mathbf{z}_{\text{OTS}}$ and the new context tokens $\mathbf{z}_{\text{CTP}}$ as a way to obtain new features $\tilde{z}_i$, which is formulated as follows:

$$\tilde{\mathbf{z}}_i = \mathbf{z}_{\text{OTS}} + \mathbf{z}_{\text{CTP}}. \tag{25}$$

### C. Mamba Layer

The mamba layer is detailed in the upper right part of Fig. 1. The augmented feature $\tilde{\mathbf{z}}_i$ is obtained from the DTA module. First, this feature is normalized and then projected into two features as follows:

$$\mathbf{s}, \mathbf{u} \leftarrow \text{Linear}\left(\text{Norm}\left(\tilde{\mathbf{z}}_i\right)\right). \tag{26}$$

Next, $\mathbf{s}$ is copied and reversed to form two directional features $\mathbf{s}_f$ and $\mathbf{s}_b$, where $f$ denotes the forward feature and $b$ denotes the backward feature. Both $\mathbf{s}_f$ and $\mathbf{s}_b$ are then processed simultaneously by 1-D convolution Conv1d$(\cdot)$ and SiLU$(\cdot)$ activation function

$$\bar{\mathbf{s}}_f = \text{SiLU}\left(\text{Conv1d}_f\left(\mathbf{s}_f\right)\right) \tag{27}$$

$$\bar{\mathbf{s}}_b = \text{SiLU}(\text{Conv1d}_b(\mathbf{s}_b)). \tag{28}$$

The processed features are then input into the mamba model

$$\tilde{\mathbf{s}}_f = \text{Mamba}_f\left(\bar{\mathbf{s}}_f\right) \tag{29}$$

$$\tilde{\mathbf{s}}_b = \text{Mamba}_b\left(\bar{\mathbf{s}}_b\right). \tag{30}$$

On this basis, the activated feature $\mathbf{u}$ is used for feature selection on the output of the mamba model

$$\hat{\mathbf{s}}_f = \tilde{\mathbf{s}}_f \otimes \text{SiLU}(\mathbf{u}) \tag{31}$$

$$\hat{\mathbf{s}}_b = \tilde{\mathbf{s}}_b \otimes \text{SiLU}(\mathbf{u}) \tag{32}$$

where $\otimes$ denotes Hadamard multiplication. Finally fusing the forward and backward outputs and adding the jump connections of the initial features, which operates as follows:

$$\mathbf{o}_i = \text{Linear}\left(\hat{\mathbf{s}}_f + \hat{\mathbf{s}}_b\right) + \tilde{\mathbf{z}}_i. \tag{33}$$

### D. Loss of Classified Compensation

The predicted probability $\mathbf{p}_i$ of the sample $\mathbf{x}_i$ is obtained using the classification head $\text{H}_{\text{cls}}$ after multiple feature augmentation and extraction. We consider that even if the original features are augmented, the range of features learned by the model corresponding to the respective class shrinks to a small extent after extensive training. This may bias the model toward a particular domain and prevent good generalization over the target domain. Drawing inspiration from the work of Mixup [42], we utilize the loss of classified compensation $\mathcal{L}_{\text{CC}}$ to suppress the overfitting of the model to a particular domain. In the DTA module, we use random samples to make context tokens that may introduce random sample object information even after perturbing the spatial relations and spectral information of the random sample tokens. Therefore, we utilize both the labels of the original samples and the random samples to guide the model in predicting the augmented features. This will effectively inhibit the model from learning features that narrow down the range of features corresponding to their respective classes to a small extent. The ablation experiments in Section IV demonstrate the positive effect of $\mathcal{L}_{\text{CC}}$ in improving the generalization ability of the model. Specifically, we first compute the cross-entropy loss $\mathcal{L}_{ce}$ using the label $\{\mathbf{y}_i\}_{i=1}^B$ of the original samples $\{\mathbf{x}_i\}_{i=1}^B$

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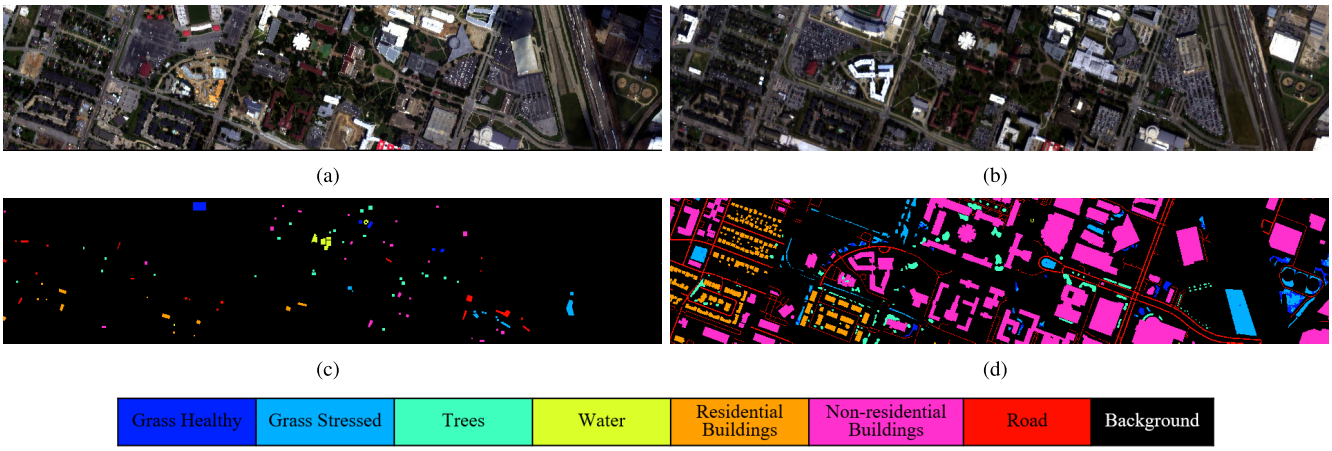



Fig. 2. Pseudo-color images and ground-truth maps of the Houston dataset. (a) Pseudo-color image of Houston 2013. (b) Pseudo-color image of Houston 2018. (c) Ground-truth map of Houston 2013. (d) Ground-truth map of Houston 2018.

and the label $\{\mathbf{y}_j\}_{i=1}^{B^s}$ of the random samples of $\{\mathbf{x}_j\}_{i=1}^{B^s}$ with predicted probability $\{\mathbf{p}_i\}_{i=1}^{B}$, respectively

$$\mathcal{L}_{\text{ori}} = \frac{1}{B}\sum_{i}\mathcal{L}_{ce}\left(\mathbf{p}_i, \mathbf{y}_i\right) \tag{34}$$

$$\mathcal{L}_{\text{ran}} = \frac{1}{B^s}\sum_{j=i}\mathcal{L}_{ce}\left(\mathbf{p}_i, \mathbf{y}_j\right) \tag{35}$$

where $B$ denotes the batch sequence of the original sample, and $B^s$ denotes the batch sequence of the random sample. The loss $\mathcal{L}_{ce}$ is formulated as follows:

$$\mathcal{L}_{ce}(\mathbf{p}, \mathbf{y}) = -\sum_{c} y^c \log p^c \tag{36}$$

where $c$ denotes the index of the class. Finally, we linearly combine $\mathcal{L}_{\text{ori}}$ and $\mathcal{L}_{\text{ran}}$ using the same ratio as the token selection ratio $\alpha$ in the DTA module to obtain the final loss $\mathcal{L}_{\text{CC}}$ as follows:

$$\mathcal{L}_{\text{CC}} = \alpha\mathcal{L}_{\text{ori}} + (1-\alpha)\mathcal{L}_{\text{ran}}. \tag{37}$$

In the testing phase, only the mamba layer and the classification head are used for feature extraction and class prediction of the target domain samples.

## IV. Experimental Results and Discussion

In this section, we utilize three cross-scene HSI datasets containing the Houston dataset, the Pavia dataset, and the HyRank dataset to assess the capability of the proposed DTAM for cross-scene classification of HSI.

### A. Experimental Data

*1) Houston Dataset:* The dataset comprises scenes from Houston 2013 [43] and Houston 2018 [44]. These scene images were captured at different times within and around the University of Houston campus using the ITRES CASI-1500H sensor. The spatial dimension of the Houston 2013 scene image is 610 × 340 pixels in size and also contains 144 spectral bands covering 380–1050 nm. The original Houston 2018 scene image has the same wavelength coverage as Houston 2013, but only 48 spectral bands. The two scene images contain the same seven classes. To fulfill the experimental requirements, we only kept the same 48 spectral bands of the two scene images and selected the overlapping spatial region of 209 × 955 pixels for the two scenes. Detailed information about the classes and sample sizes of the two scene image is shown in Table I, and their ground-truth maps and pseudo-color images are shown in Fig. 2. The raw data for the Houston dataset was obtained from https://hyperspectral.ee.uh.edu/?page_id=29.

TABLE I
Number of Source and Target Samples for the Houston Dataset

| No. | Name | Houston 2013 | Houston 2018 |
|---|---|---|---|
| 1 | Grass Healthy | 345 | 1353 |
| 2 | Grass Stressed | 365 | 4888 |
| 3 | Trees | 365 | 2766 |
| 4 | Water | 285 | 22 |
| 5 | Residential Buildings | 319 | 5347 |
| 6 | Non-residential Buildings | 408 | 32459 |
| 7 | Road | 443 | 6365 |
| | Total | 2530 | 53200 |

*2) Pavia Dataset:* The Pavia dataset consists of the University of Pavia (UP) scene and the Pavia Center (PC) scene. These scene images were taken at different times over the Pavia region using the reflective optics system imaging spectrometer (ROSIS) sensor. The spatial dimension of the UP scene image is 610 × 430 pixels and also contains 102 spectral bands covering 430–860 nm. The two scene image have the same seven classes. We deleted the last band in the UP scene image to fulfill the experimental requirements. Detailed information about the classes and the number of samples for the two scene images is provided in Table II. Their ground-truth maps and pseudo-color images are displayed in Fig. 3. The raw data for the Pavia dataset was obtained from https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Center_and_University.

*3) HyRank Dataset:* The HyRank dataset contains the Dioni and Loukia scenes. These two scene images were captured by the Hyperion sensor (EO-1, USGS). The spatial dimension

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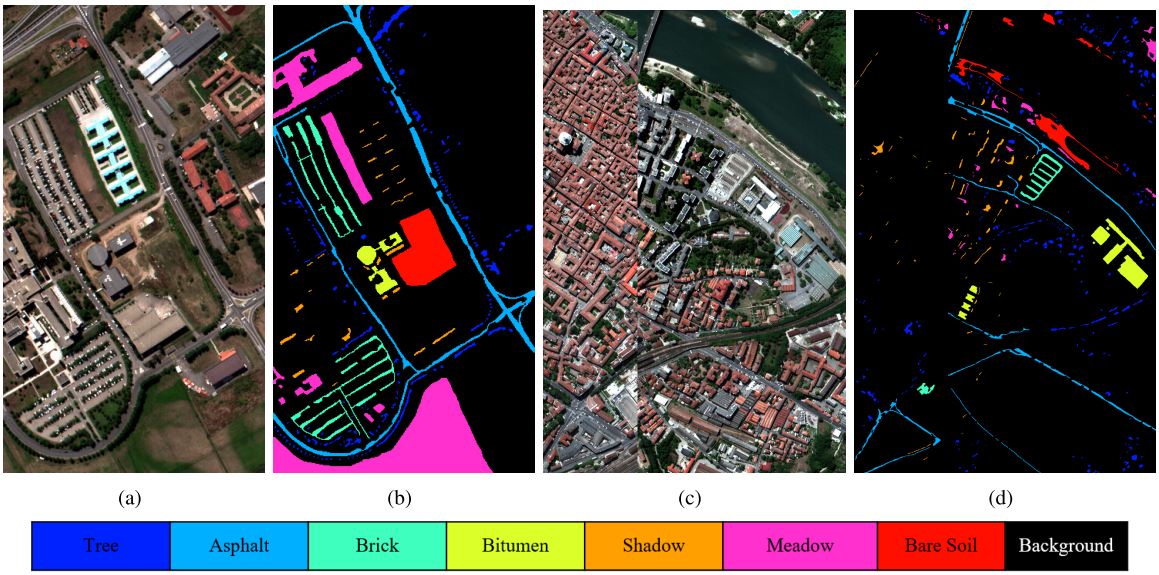



Fig. 3. Pseudo-color images and ground-truth maps of the Pavia dataset. (a) Pseudo-color image of UP. (b) Ground-truth map of UP. (c) Pseudo-color image of PC. (d) Ground-truth map of PC.

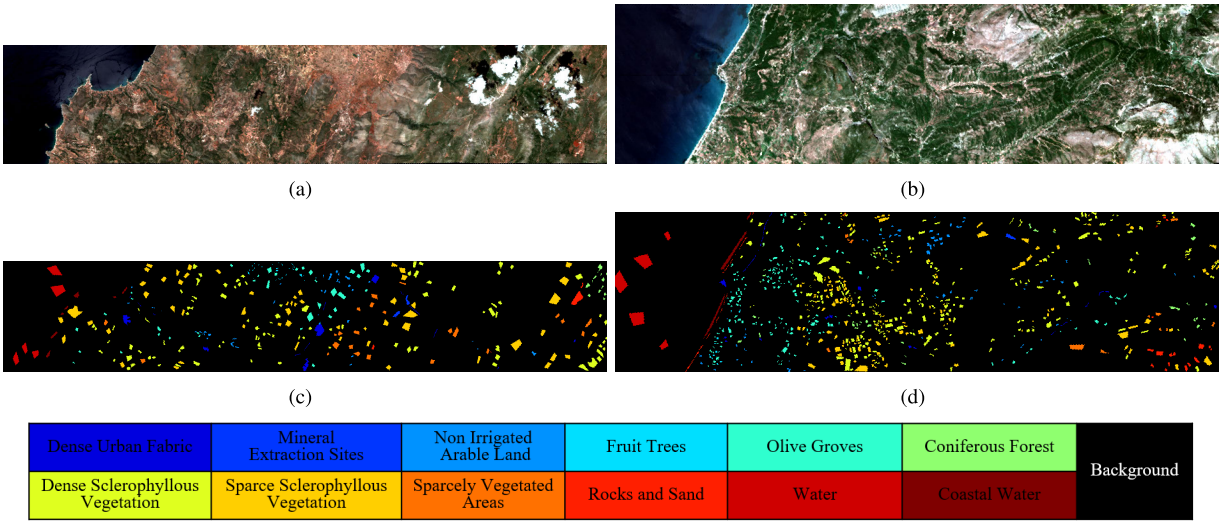



Fig. 4. Pseudo-color images and ground-truth maps of the HyRank dataset. (a) Pseudo-color image of Dioni. (b) Pseudo-color image of Loukia. (c) Ground-truth map of Dioni. (d) Ground-truth map of Loukia.

of the Dioni scene image is 250 × 1376 pixels, while the Loukia scene image measures 249 × 945 pixels. The two scenes contain 176 of the same spectral bands and the same 12 classes. Detailed information about the classes and number of samples of the image for the two scenes is provided in Table III. Their ground-truth maps and pseudo-color images are displayed in Fig. 4. The raw data for the HyRank dataset was obtained from https://zenodo.org/records/1222202.

### B. Experimental Setting

We set up three cross-scene classification tasks of HSI throughout the whole experiment, which are called the Houston task, Pavia task, and HyRank task. In the Houston task, Houston 2013 scene is used as SD and Houston 2018 scene is used as TD. In the Pavia task, UP scene is used as SD, and PC scene is used as TD. In the HyRank task, Dioni scene is used as SD, and Loukia scene is used as TD. In all tasks, the training samples were selected only for labeled SD (80% for training and 20% for validation). We used the accuracy of each class, the overall accuracy (OA), and the Kappa coefficient (kappa) as evaluation metrics for the experiments. The test results

TABLE II
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE PAVIA DATASET

| No. | Name | UP | PC |
|---|---|---|---|
| 1 | Tree | 3064 | 7598 |
| 2 | Asphalt | 6631 | 9248 |
| 3 | Brick | 3682 | 2685 |
| 4 | Bitumen | 1330 | 7287 |
| 5 | Shadow | 947 | 2863 |
| 6 | Meadow | 18649 | 3090 |
| 7 | Bare Soil | 5029 | 6584 |
| | Total | 39332 | 39355 |

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TABLE III
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HyRank DATASET

| No. | Name | Dioni | Loukia |
|---|---|---|---|
| 1 | Dense Urban Fabric | 1262 | 288 |
| 2 | Mineral Extraction Sites | 204 | 67 |
| 3 | Non Irrigated Arable Land | 614 | 542 |
| 4 | Fruit Trees | 150 | 79 |
| 5 | Olive Groves | 1768 | 1401 |
| 6 | Coniferous Forest | 361 | 500 |
| 7 | Dense Sclerophyllous Vegetation | 5035 | 3793 |
| 8 | Sparce Sclerophyllous Vegetation | 6374 | 2803 |
| 9 | Sparsely Vegetated Areas | 1754 | 404 |
| 10 | Rocks and Sand | 492 | 487 |
| 11 | Water | 1612 | 1393 |
| 12 | Coastal Water | 398 | 451 |
| | Total | 20024 | 12208 |

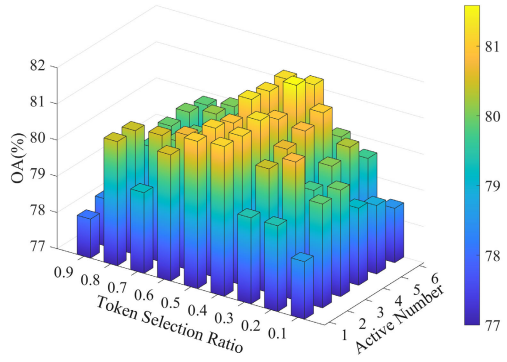



Fig. 5. Hyperparameter tuning of token selection ratio and active number for the proposed DTAM on the Houston task.

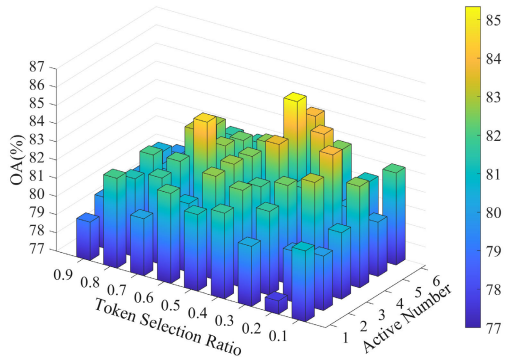



Fig. 6. Hyperparameter tuning of token selection ratio and active number for the proposed DTAM on the Pavia task.

were expressed as mean ± standard deviation by repeating the experiment five times with different random seeds selected to assess the stability of the proposed DTAM. Additionally, note that we mainly explore the ability of the DTA module and the classification compensation loss to enhance the model generalization effect, thus the mamba layer for all the following experimental DTAMs is set to six layers.

The comparison algorithms employed in the comparative analysis experiments include style-agnostic networks (SagNets) [45], progressive domain expansion network (PDEN) [26], domain shifts with uncertainty (DSU) [46], AdaptivelyMix [47], single-source domain expansion network (SDEnet) [28], language-aware DG network (LDGnet) [48], locally linear unbiased randomization network (LLURnet) [27], and spatial–spectral enhancement and causal constraint network (S2ECNet) [29]. The SagNets, DSU, AdaptivelyMix, and PDEN algorithms are designed for the natural image DG field. Therefore, their input ports need to be fine-tuned so that samples from HSIs can be input properly. To guarantee fairness in comparison, all methods were developed using the PyTorch platform and trained from the ground up on an Nvidia RTX 3090 GPU with 24 GB of memory. Moreover, each method was optimized with adaptive moment estimation (Adam), utilizing a momentum of 0.9 and an initial learning rate of 0.001. In the Houston task, the number of SD samples was increased fourfold using random flip and random radiation noise augmentation techniques. The parameters for all tasks remain constant: batch size (256), patch size (13), training epochs (200), and L2 decay (0.0005).

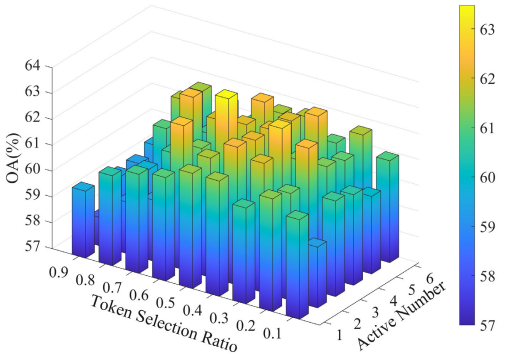



Fig. 7. Hyperparameter tuning of token selection ratio and active number for the proposed DTAM on the HyRank task.

### C. Parameter Tuning

We performed parameter analysis experiments to explore the optimal performance of the DTAM method on three cross-scene classification tasks and analyzed the effect of different parameters on the performance of the method. In this method, the token selection ratio, the active number of the DTA module, and the embedding dimension of the features as adjustable hyperparameters are selected from {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9}, {1, 2, 3, 4, 5, 6}, and {32, 64, 128, 256, 512}, respectively.

The token selection ratio and the active number of DTA modules are important parameters that affect the generalization performance of the DTAM method. Therefore, we use cross-validation to verify the degree to which these two hyperparameters affect the method. Figs. 5–7 show the trend of the performance of the proposed DTAM method on three cross-scene classification tasks of HSI under different combinations of token selection ratio and active number. We observe that the DTAM method has higher classification accuracy on the three tasks when the token selection ratio is taken as 0.4 or 0.5. The token selection ratio controls the preservation ratio of object information in the original sample features; the larger the value, the more original sample object tokens

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TABLE IV
ABLATION EXPERIMENTS WITH VARIOUS VARIANTS OF DTAM

| Model | DTAM (no_OTS) | DTAM (no_CTP) | DTAM (no_DTA) | DTAM (CEloss) | DTAM (ViT) | DTAM |
|---|---|---|---|---|---|---|
| Task | Houston | | | | | |
| OA(%) | 77.09±1.64 | 78.13±1.20 | 74.45±1.04 | 79.58±0.97 | 78.91±1.42 | **81.60±0.83** |
| kappa(×100) | 62.81±2.24 | 63.98±1.68 | 58.43±1.55 | 66.50±1.41 | 64.47±1.19 | **68.60±0.89** |
| Task | Pavia | | | | | |
| OA(%) | 81.36±2.61 | 81.65±2.97 | 80.14±1.68 | 84.38±2.33 | 78.73±1.31 | **84.03±1.18** |
| kappa(×100) | 79.82±4.59 | 79.02±4.65 | 77.32±3.99 | 82.08±3.27 | 75.65±3.27 | **81.62±1.40** |
| Task | HyRank | | | | | |
| OA(%) | 61.32±0.94 | 58.28±2.31 | 55.60±2.33 | 61.16±0.72 | 60.08±0.95 | **62.88±0.54** |
| kappa(×100) | 51.04±1.61 | 46.44±5.11 | 42.91±3.69 | 50.80±2.12 | 47.98±2.67 | **53.45±1.11** |

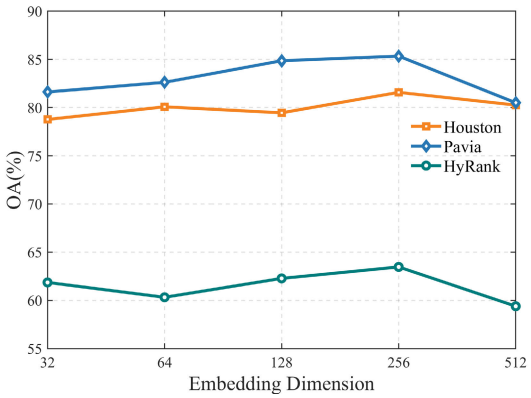



Fig. 8. Sensitivity analysis of embedding dimension parameters.

are preserved. However, it does not mean that a higher token selection ratio is better. A higher token selection ratio will keep the contextual information of the original features, which is not good for the generalization of the model. A lower token selection ratio will cause some object information corresponding to the class label to be lost, which makes the model's classification accuracy decrease. Meanwhile, the proposed DTAM method achieves the highest performance when the parameter active number is taken as 5, 5, and 3, respectively. The embedding dimension of features affects the ability of the mamba layer to extract features and the amount of information contained in the features, respectively. Fig. 8 presents the variation in the performance of the proposed DTAM method with the change in the feature embedding dimension parameter. When the embedding dimensions of the features are 256, the DTAM method achieves optimal accuracy in the three cross-scene classification tasks.

### D. Ablation Study

The OTS submodule and the CTP submodule are the key components of the DTA module. The loss of classified compensation is the main strategy of the optimization model. To assess the effectiveness of key modules in the DTAM method, we performed an ablation experiment analysis by removing significant parts from the entire network. Table IV presents the results of each variant of the DTAM method on three cross-scene classification tasks of HSI which contain five variants: 1) "DTAM (no_OTS)": the OTS submodule is removed from the network; 2) "DTAM (no_CTP)": the CTP submodule is removed from the network; 3) "DTAM (no_DTA)": the DTA module is removed from the network; 4) "DTAM (CEloss)": the loss of classified compensation is removed from the network; and 5) "DTAM (ViT)": the mamba layer in the network is replaced using the vision transformer.

The results in Table IV highlight the superiority of the proposed DTAM over other variants, demonstrating that the absence of any of the components leads to a decrease in the generalization ability of the method. Specifically, the variants lacking DTA had the largest decreases on three tasks, with decreases of 7.15%, 3.89%, and 7.28% compared to the DTAM method on the Houston task, the Pavia task, and the HyRank task, respectively. This result demonstrates the effectiveness of the DTA module for feature augmentation. Second, the lack of the OTS submodule or a CTP submodule also reduces the ability of the method for cross-scene classification. Experimental results with the "DTAM (CEloss)" variant demonstrate that the loss of classified compensation is also a key part of enhancing the model's generalization ability. The variant that lacked loss of classified compensation decreased classification accuracy by 1%–3% on the three tasks. The results of the variant "DTAM (ViT)" on the three tasks indicate that the mamba has higher robustness compared to the vision transformer in capturing global features of HSIs. Particularly, in the Pavia task, the OA of the variant "DTAM (ViT)" is only 78.73%, which is decreased by 5.3% compared to the DTAM method.

### E. Comparison and Analysis

To compare the performance of DTAM, eight methods are selected as comparison methods for this experiment. Among them, SagNets, PDEN, DSU, and AdaptiveMix are methods designed for single DG of natural images. SDEnet, LDGnet, LLURnet, and S2ECNet are designed for cross-scene classification of HSI. Tables V–VII present the performance of the eight compared methods and the proposed DTAM method on three cross-scene classification of HSI tasks. The top-performing results are highlighted in bold.

From the experimental results, the classification accuracies of the four methods for natural image generalization are

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TABLE V
EACH CLASS AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT ALGORITHMS ON THE HOUSTON TASK

| Class | SagNets [45] | PDEN [26] | DSU [46] | AdaptiveMix [47] | SDEnet [28] | LDGnet [48] | LLURnet [27] | S2ECNet [29] | DTAM |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 24.86 | 66.67 | 39.67 | 11.80 | 45.37 | 64.70 | 32.91 | **74.12** | 51.06 |
| 2 | 67.63 | 82.56 | 78.40 | 77.57 | 73.26 | 78.05 | 79.64 | 75.46 | **88.30** |
| 3 | 59.81 | 54.63 | 59.59 | 65.69 | 60.78 | 54.71 | 65.93 | **74.82** | 60.93 |
| 4 | 80.30 | **100.00** | **100.00** | 80.91 | **100.00** | 73.64 | **100.00** | **100.00** | 96.36 |
| 5 | 57.43 | 69.38 | 65.76 | 67.47 | 58.90 | 67.53 | 56.30 | 64.83 | **81.60** |
| 6 | 70.34 | 76.49 | 88.09 | **94.03** | 89.07 | 90.34 | 91.05 | 84.87 | 90.25 |
| 7 | 57.28 | **59.45** | 58.34 | 0.48 | 57.12 | 44.70 | 44.10 | 46.63 | 47.79 |
| OA(%) | 65.53±0.90 | 72.91±0.37 | 78.69±1.06 | 75.09±0.17 | 78.18±0.35 | 78.95±0.48 | 78.11±0.94 | 76.63±1.13 | **81.60±0.83** |
| kappa(×100) | 47.53±0.60 | 57.55±0.38 | 63.89±2.18 | 52.83±0.48 | 62.26±1.64 | 63.67±0.88 | 61.22±0.71 | 61.48±1.14 | **68.60±0.89** |

TABLE VI
EACH CLASS AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT ALGORITHMS ON THE PAVIA TASK

| Class | SagNets [45] | PDEN [26] | DSU [46] | AdaptiveMix [47] | SDEnet [28] | LDGnet [48] | LLURnet [27] | S2ECNet [29] | DTAM |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 91.27 | 87.79 | 93.67 | 87.65 | 88.58 | **96.63** | 77.16 | 85.23 | 94.21 |
| 2 | 79.82 | 84.58 | 83.61 | 83.22 | 78.60 | 85.94 | 85.27 | 87.13 | **87.82** |
| 3 | 38.08 | 49.4 | 19.91 | 65.47 | 70.96 | 52.79 | **76.33** | 75.85 | 60.39 |
| 4 | 76.01 | **87.51** | 69.25 | 75.31 | 85.65 | 79.29 | 84.51 | 82.15 | 84.48 |
| 5 | 74.52 | 86.03 | 93.41 | 82.88 | **93.47** | 91.86 | 81.07 | 79.65 | 88.81 |
| 6 | 74.13 | 82.52 | 65.44 | **82.91** | 76.43 | 63.49 | 79.95 | 71.81 | 73.15 |
| 7 | 53.70 | 70.43 | 73.87 | 62.50 | 75.71 | 82.10 | **89.63** | 77.49 | 79.11 |
| OA(%) | 73.27±0.82 | 80.92±1.15 | 76.20±1.05 | 77.88±0.65 | 81.74±1.08 | 82.54±1.82 | 82.97±1.04 | 81.71±0.59 | **84.03±1.18** |
| kappa(×100) | 68.20±1.05 | 77.23±1.52 | 71.69±1.96 | 74.09±0.96 | 78.32±1.35 | 79.08±2.52 | 80.38±1.21 | 79.00±0.71 | **81.62±1.40** |

TABLE VII
EACH CLASS AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT ALGORITHMS ON THE HyRank TASK

| Class | SagNets [45] | PDEN [26] | DSU [46] | AdaptiveMix [47] | SDEnet [28] | LDGnet [48] | LLURnet [27] | S2ECNet [29] | DTAM |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 32.64 | 29.79 | 28.82 | 15.35 | 13.26 | **47.71** | 21.53 | 12.85 | 11.6 |
| 2 | 0.00 | 8.95 | 6.86 | 1.19 | 6.87 | 20.00 | 56.12 | 60.60 | **86.86** |
| 3 | 37.42 | 29.96 | 42.10 | 11.92 | 33.21 | **57.20** | 33.03 | 4.91 | 3.50 |
| 4 | 2.78 | 0.00 | 11.39 | 6.58 | 9.11 | **21.01** | 8.10 | 7.34 | 4.05 |
| 5 | 4.23 | 2.58 | **26.82** | 15.18 | 12.03 | 0.54 | 3.67 | 25.87 | 21.23 |
| 6 | 18.84 | 25.72 | 9.68 | 2.92 | 26.8 | **35.92** | 27.04 | 4.00 | 21.12 |
| 7 | 71.89 | 80.79 | 72.46 | 74.94 | 78.37 | 81.96 | 82.64 | **85.83** | 78.73 |
| 8 | 65.68 | 64.72 | 67.61 | 51.82 | 63.13 | 47.52 | 65.66 | 58.26 | **68.72** |
| 9 | 17.87 | 45.59 | 57.62 | **70.00** | 68.47 | 69.7 | 48.81 | 69.21 | 65.29 |
| 10 | 20.54 | 0.16 | 0.45 | 5.95 | 3.70 | 3.00 | 2.22 | 8.38 | **35.9** |
| 11 | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | **100.00** | 99.93 | **100.00** |
| 12 | **100.00** | **100.00** | **100.00** | **100.00** | 99.69 | **100.00** | **100.00** | 99.91 | 99.29 |
| OA(%) | 58.43±0.88 | 60.01±1.38 | 61.19±0.90 | 55.64±1.20 | 60.71±0.77 | 59.35±0.60 | 61.43±0.73 | 61.79±0.79 | **62.88±0.54** |
| kappa(×100) | 47.28±0.52 | 49.38±1.38 | 51.48±1.56 | 44.49±1.33 | 50.73±0.78 | 49.49±0.72 | 51.07±0.87 | 51.71±1.46 | **53.45±1.11** |

overall lower than those for cross-scene classification of HSI. Particularly, in the Pavia task, the OA of PDEN, which is the best-performing DG method for natural images, is only 80.92%, which is 0.79% lower than S2ECNet and 3.11% lower than the proposed DTAM. Additionally, the methods for DG of natural images have unstable performance on different cross-scene classification tasks. Across the three tasks, the DTAM method achieved the highest results on the OA and kappa evaluation metrics compared to the eight comparison methods. On the Houston task, the OA of DTAM scored 81.60%, outpacing second-place LDGnet by 2.65%. On the Pavia task, the OA of DTAM achieved 84.03%, outperforming the second-place LDGnet by 1.06%. On the HyRank task, the OA of DTAM achieved 62.88%, outperforming the second-place S2ECNet by 1.09%. Among them, for the recognition of the tenth class of rocks and sand in the Loukia scene, the correct recognition rate of other methods is very low, while the correct recognition rate of DTAM reaches 35.9%.

To visually assess DTAM and the eight compared methods qualitatively, Figs. 9–11 show a visualization of the classification results of all methods on the three tasks. The classification maps from DTAM are closer to the ground truth on all three target scenes than the maps from the comparison methods. Most notably, at the lower left in Fig. 11, only the DTAM method correctly classifies rocks and sand as red, while the other methods misclassify them as orange in the ninth class of sparsely vegetated areas.

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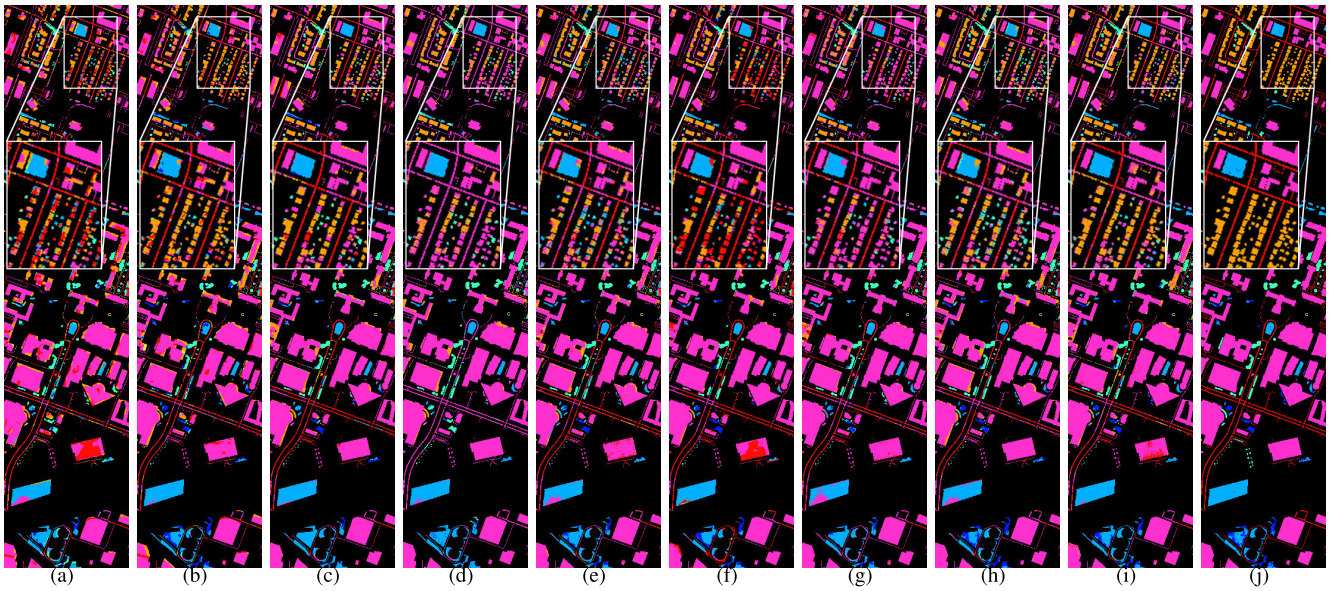



Fig. 9. Visual representation of classification results on the Houston task obtained with different methods. (a) SagNets (68.02%). (b) PDEN (73.17%). (c) DSU (79.89%). (d) AdaptiveMix (75.27%). (e) SDEnet (78.52%). (f) LDGnet (79.59%). (g) LLURnet (77.97%). (h) S2ECNet (78.08%). (i) DTAM (81.58%). (j) Ground truth.

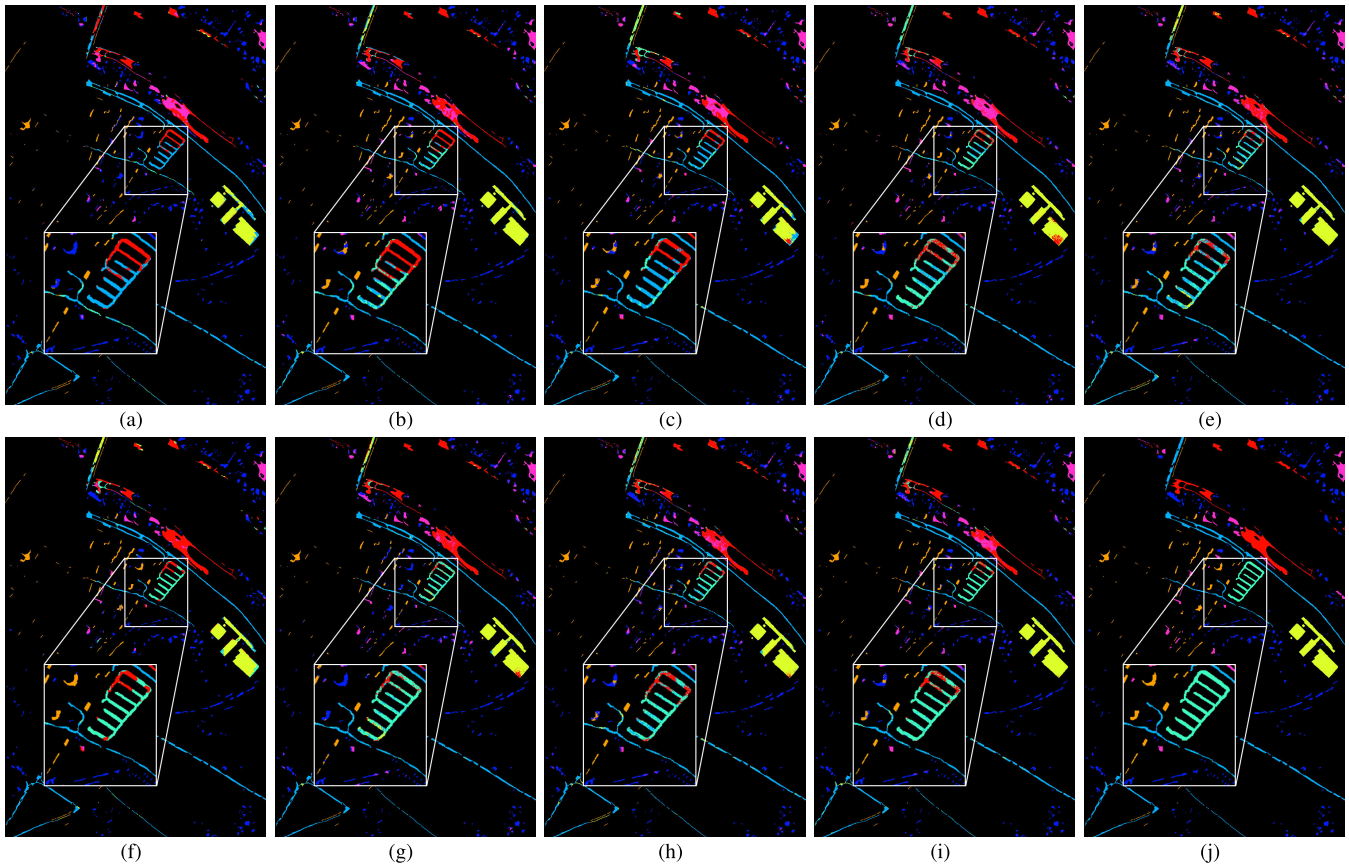



Fig. 10. Visual representation of classification results on the Pavia task obtained with different methods. (a) SagNets (73.21%). (b) PDEN (81.82%). (c) DSU (76.9%). (d) AdaptiveMix (78.82%). (e) SDEnet (82.26%). (f) LDGnet (84.49%). (g) LLURnet (84.02%). (h) S2ECNet (82.15%). (i) DTAM (85.34%). (j) Ground truth.

Table VIII presents the training times for all methods on the three tasks with the same batch size of 256. The training time for the DTAM method is somewhat longer compared to the computation time of the comparison method using CNN as the backbone network, due to DTAM employing mamba as its backbone. However, for all three tasks, the training time for DTAM was kept under one minute.

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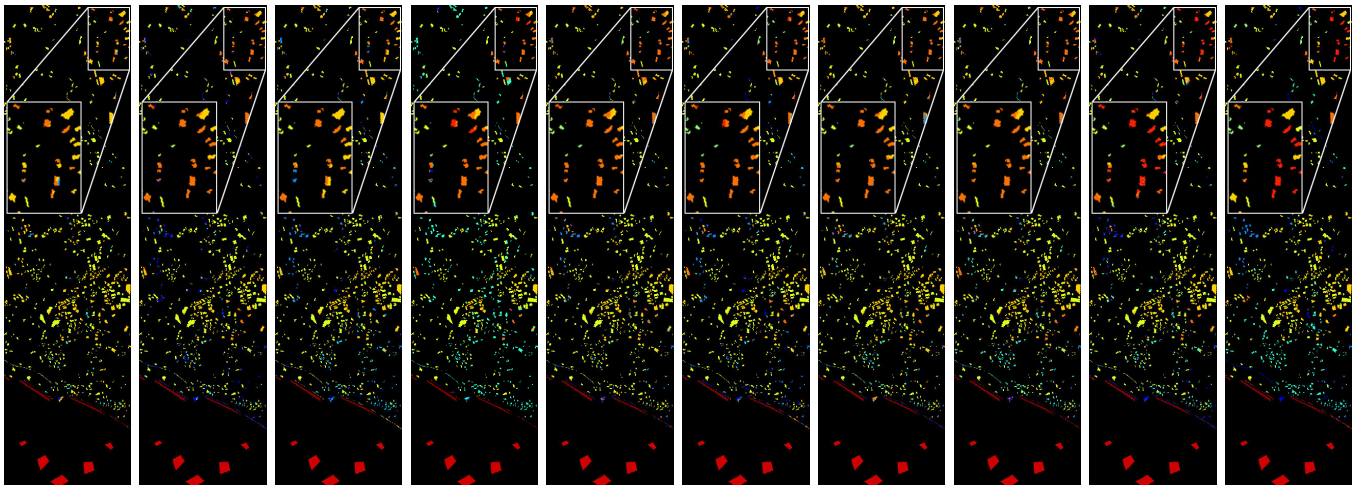



Fig. 11. Visual representation of classification results on the HyRank task obtained with different methods. (a) SagNets (59.73%). (b) PDEN (60.03%). (c) DSU (61.45%). (d) AdaptiveMix (55.46%). (e) SDEnet (60.98%). (f) LDGnet (60.38%). (g) LLURnet (62.02%). (h) S2ECNet (62.05%). (i) DTAM (63.47%). (j) Ground truth.

TABLE VIII
TRAINING TIME (IN SECONDS) OF ONE EPOCH IN DIFFERENT METHODS ON THREE TASKS

| Task | SagNets [45] | PDEN [26] | DSU [46] | AdaptiveMix [47] | SDEnet [28] | LDGnet [48] | LLURnet [27] | S2ECNet [29] | DTAM |
|---|---|---|---|---|---|---|---|---|---|
| Houston | 7.77 | 4.66 | 4.57 | 8.99 | 6.21 | 30.78 | 13.69 | 26.48 | 12.33 |
| Pavia | 19.62 | 13.16 | 23.72 | 63.06 | 33.66 | 101.57 | 26.64 | 73.84 | 46.55 |
| HyRank | 11.09 | 9.97 | 18.97 | 43.33 | 36.51 | 74.56 | 21.03 | 42.66 | 29.01 |

## V. CONCLUSION

In this article, we design a novel method for the cross-scene classification of HSI with the backbone network using mamba, called DTAM. The DTA module is the method's core. It mainly consists of the OTS submodule and the CTP submodule. The OTS implements the preservation of tokens with object information in the original features. The CTP makes context tokens with more complex information using random sample features. The original features are implicitly augmented by reorganizing the outputs of the two submodules. Meanwhile, we utilize the loss of classified compensation to enhance the robustness of the features learned by the model. Comprehensive extensive experiments conducted on three cross-scene classification tasks of HSI demonstrate the effectiveness and superiority of the proposed DTAM method. In future work, we will explore more direct and effective ways of token-level spectral randomization to further reduce the impact of domain differences on cross-scene classification of HSI. In addition, we will introduce detailed sample description information to assist the model in capturing image domain-invariant features.

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